

lasso cells or **colloblasts** which help in food capture. Digestion is both extracellular and intracellular. **Bioluminescence** (the property of emitting light by living organism) is well-marked in the ctenophores. Sexes are not separate (monoecious). Reproduction takes place only by sexual method. Fertilisation is external. Development is indirect and includes a larval stage called **cydippid larva**.

Examples: *Pleurobrachia*, *Hormiphora*.

3.3.2 Classification of the phylum Ctenophora

Phylum Ctenophora is classified into two classes,

1. Tentaculata and 2. Nuda.

Class-1: Tentaculata

The adults possess two aboral tentacles.

Examples: *Pleurobrachia*, *Ctenoplana*.

Class-2: Nuda

Tentacles are absent even in the larval stages.

Examples: *Beroë*.

Note: Sea anemones and sea walnuts are said to be 'Biradially Symmetrical'.

3.4 PHYLUM-PLATYHELMINTHES

- *Mesoderm and the bilateral plan*

3.4.1 General Characters

They have **dorso-ventrally flattened body**, hence are called **flatworms**. These are mostly endoparasites found in animals, including human beings. Flatworms are the first **bilaterally symmetrical**, **triploblastic** and **acoelomate** animals with **organ-system level of organisation** (Fig. 3.6-3.8). Body is not segmented, but some exhibit pseudometamerism. They show moderate cephalization and unidirectional movement (locomotion). **Hooks** and **suckers** are present in the parasitic forms (Fig. 3.5). Digestive system, if present, has mouth only; anus is absent. Nervous system is moderately developed with brain (cephalic ganglia) and nerve cords forming 'ladder like' system. Specialised excretory cells called **flame cells (protonephridia)** help in osmoregulation and excretion. Sexes are not separate (monoecious). Fertilisation is internal and development is indirect with many stages (**miracidium**, **sporocyst**, **redia**, **cercaria** etc.). **Polyembryony** is common in some (Liver

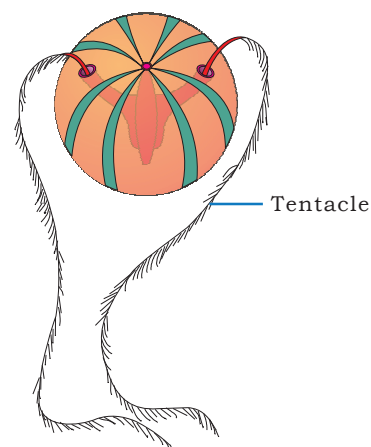


Fig. 3.4 Example of ctenophora (*Pleurobrachia*)

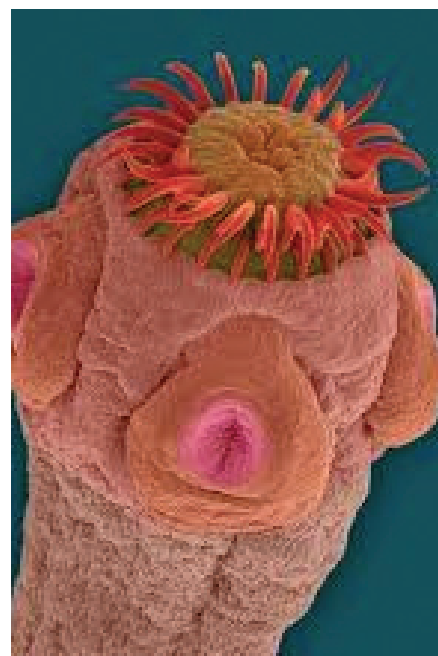
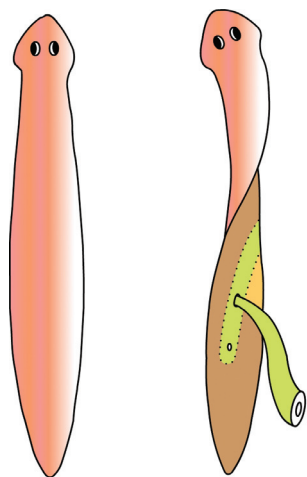
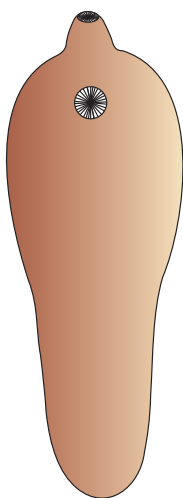
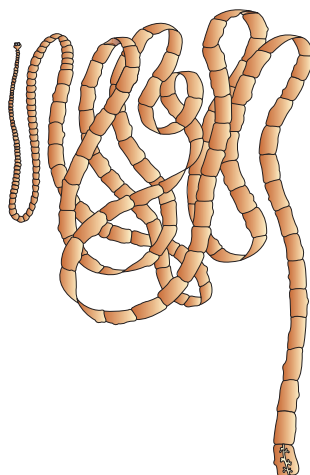


Fig. 3.5 Scolex of tape worm

Fig. 3.6 *Dugesia*Fig. 3.7 *Fasciola*Fig. 3.8 *Taenia*

fluke). Some members like *Planaria* possess high degree of **regeneration** capacity.

Examples: *Taenia*, *Fasciola*, *Schistosoma*, *Dugesia*.

3.4.2 Classification of the phylum Platyhelminthes

Phylum Platyhelminthes is classified into three classes, namely **1. Turbellaria** **2. Trematoda** and **3. Cestoda**.

Class-1: Turbellaria

They are mostly free living, unsegmented flat worms. Body wall bears cilia and mucus forming structures called **rhabdites** in the epidermal cells. Reproduction is by asexual and sexual methods. Development is indirect and includes free swimming, ciliated **Muller's larva**.

Examples: *Dugesia*, *Convoluta*.

Class-2: Trematoda

They are commonly called *flukes*. They are parasitic on other animals. Body is covered by a thick cuticle (tegument); bears two suckers, an oral and a ventral (acetabulum). Mouth is anterior and the intestine is bifurcated. They are bisexual (monoecious). Life history is complex with many hosts and different types of stages (*miracidium*, *sporocyst*, *redia*, *cercaria* etc.).

Examples: *Fasciola* (liver fluke), *Schistosoma* or *Bilharzia* (blood fluke).

Class-3: Cestoda

They are commonly called *tapeworms*. All are endoparasites. Body is covered by protective **syncytial tegument**. They show **pseudometamerism** (body is divided into segment like units called proglottids which are not 'true segments'). Scolex bears hooks, suckers and **bothridia** (leaf like outgrowths) for anchoring. Mouth and gastrovascular cavity are absent. They are bisexual (monoecious). Life history is complex with many hosts and development is indirect with cysticercus larva (bladder worm).

Examples: *Taenia solium* (pork tapeworm), *Echinococcus granulosus* (dog tapeworm).